Department of Artificial Intelligence and Machine Learning

Course Code : **AI362IA** Course : **Big Data Technologies**

Project Synopsis

Project Title	**Adaptive Analysis of Web Traffic for Automated Bot Identification**
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Introduction

The rapid growth of web applications has led to a significant increase in automated traffic, including both beneficial and malicious agents. Malicious automated activity can degrade system performance, distort analytics, and pose serious security risks. Traditional detection methods such as static rules or CAPTCHA systems are often either ineffective against advanced automation or intrusive to genuine users. Therefore, there is a need for scalable and intelligent approaches that can efficiently analyze large volumes of traffic data. This project leverages Apache Spark to perform distributed analysis of web traffic, enabling efficient and accurate identification of automated activity, making it highly relevant to modern cybersecurity and large-scale data analytics.

Gaps Identified

1. Existing methods rely on limited feature sets and fail to capture complex traffic patterns.
2. Lack of scalable processing leads to inefficiency with large-scale web data.
3. Most systems provide only binary classification without risk-based insights or interpretability.

Problem Statement and Objectives

The project addresses the challenge of building a scalable and intelligent system capable of accurately identifying automated activity in large-scale web traffic datasets. The aim is to overcome limitations of existing approaches by leveraging distributed computing and multi-dimensional feature analysis.

The objectives are:

1. To design a distributed system using Apache Spark for efficient processing of large datasets.
2. To utilize multiple feature dimensions derived from traffic patterns and interactions for improved identification.
3. To develop a supervised machine learning model to classify human and automated activity.
4. To implement an adaptive risk scoring mechanism along with an analytical dashboard for intuitive interpretation of results.

Methodology

The proposed system follows a structured pipeline integrating distributed processing, machine learning, and visualization. Traffic-derived feature datasets are ingested into Apache Spark, merged, and preprocessed through cleaning and normalization to create a unified dataset. Relevant features capturing request intensity, access variability, and interaction diversity are utilized for modeling. A supervised machine learning model, such as Random Forest or Logistic Regression, is trained on labeled data to classify traffic instances. In addition to classification, an adaptive scoring mechanism assigns a probability score to each instance, enabling categorization into multiple risk levels. Exploratory data analysis and feature importance evaluation are performed to interpret model behavior. Finally, an analytical dashboard is developed using visualization tools to present traffic distribution, model predictions, and risk insights, enabling clear and interactive understanding of system outputs.

Tools Used

1. Apache Spark – Used for large-scale data processing and machine learning.
2. Apache HBase – Used for scalable storage and real-time access to large datasets.
3. Apache Superset – Used for building interactive dashboards and visualizing traffic insights.
4. GitHub – Used for version control and project management.

Expected Outcomes

- A scalable system capable of identifying automated activity in web traffic using distributed computing.
- Improved classification performance through the use of multiple feature dimensions and machine learning models.
- An adaptive risk scoring mechanism for categorizing traffic into different risk levels.
- An analytical dashboard for visualizing traffic patterns, model predictions, and key insights.
- Demonstration of the application of Big Data and intelligent analytics in real-world cybersecurity scenarios.